String Theory and M-Theory

Lecture 10: T-Duality, D-Branes, and Gauge Theory

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Companion notes curated by LazyingArt LLC with Video2Book

Chapter 1

T-Duality, D-Branes, and Gauge Theory

Overview. T-duality begins as an exact equivalence between closed strings on circles of radii (R) and α'/R. Momentum and winding exchange roles, as do the worldsheet derivatives $\partial_\tau Y$ and $\partial_\sigma Y$, and the spacetime vectors descending from $G_{\mu 5}$ and $B_{\mu 5}$. When the same operation is applied to an open string, a Neumann endpoint condition becomes a Dirichlet condition. The endpoint must then lie on a dynamical D-brane. Open strings confined to one brane produce an abelian gauge theory at low energy; strings joining a stack of branes carry two endpoint labels and produce a nonabelian gauge theory. A D-string ending on a D3-brane supplies the corresponding magnetic monopole.

1.1 Why Return to T-Duality?

We have already met T-duality, but we return to it for a reason. It was one of the clues that string theory contained much more than perturbative strings. Once its consequences for open strings were understood, it led to D-branes; D-branes in turn became a way to construct and study quantum field theories. The aim of this lecture is therefore not merely to repeat the slogan that a small circle is equivalent to a large circle. We shall rebuild the argument carefully enough to see how that slogan forces new physical objects into the theory.

The simplest setting is toroidal compactification. A one-dimensional torus is a circle. A two-dimensional torus can be represented by a parallelogram whose opposite sides are identified, and a three-dimensional torus by a parallelepiped whose opposite faces are identified. The same construction continues in higher dimensions. We shall keep only one compact spatial coordinate, called Y, and one representative noncompact coordinate, called X. Time is present but is not shown in the board diagrams.

Choose the standard convention

$$Y \sim Y + 2\pi R, \tag{1.1}$$

so that R is the radius and $2\pi R$ is the circumference. The lecture often calls the compact size simply r and suppresses factors of 2π and α'. We shall retain the conventional factors in the derivations and display the lecture units when they are useful.

For the first half of the discussion the strings are closed and oriented. An orientation is an intrinsic arrow drawn around the string. Reversing all arrows everywhere would change no experiment; what matters is the relative orientation of two strings. During splitting and joining, the arrows must join continuously. This is the first hint that a signed, conserved quantity will appear.

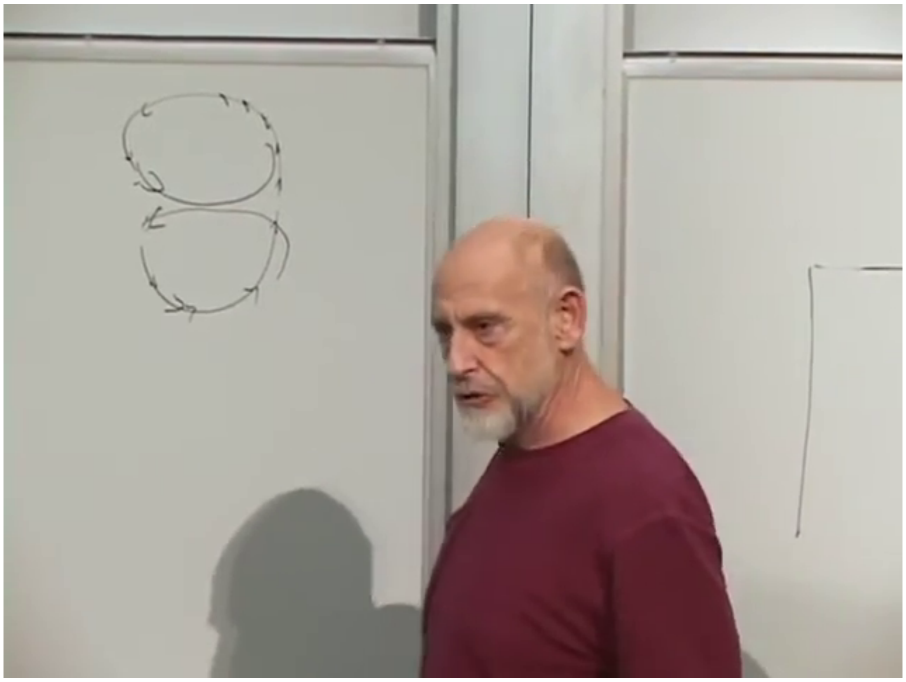

Figure 1.1: Two closed strings with their relative orientations marked on the board. Orientation must be continuous through splitting and joining.

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Question from the audience. If the worldsheet coordinate runs from $\sigma = 0$ to 2π, has the string already gone once around the compact direction? [a]

Response. No. The coordinate σ labels points along the string; it does not label positions in the compact space. Every closed string has a periodic parameter σ, whether it is wound or unwound. The rubber-band analogy is useful: one may number the molecules around a rubber band even when the band is not wrapped around anything. Winding is a property of the map $Y(\tau, \sigma)$, not of the parameter σ alone.

[a] Lecture timestamp: 00:08:42.

Question from the audience. Must the compact size also be the same at every value of the large coordinate X? [a]

Response. Not in a general dynamical compactification. We begin with a fixed circle because it is the cleanest background. Later the component G_{55} will become a scalar field whose value measures the local size of the fifth direction, so that size can vary over ordinary spacetime.

[a] Lecture timestamp: 00:10:00.

1.2 Winding Is Conserved Even When Strings Reconnect

An unwound string may move through the compact direction without wrapping it. A wound string is different: as σ goes once around the closed string, the embedding $Y(\tau,\sigma)$ advances through one or more periods of the compact circle. Because the string is oriented, the advance has a sign. A string can wind in the positive direction or in the negative direction.

The conservation law is clearest in the joining pictures. Take one string with winding $+1$ and another with winding -1. Their compatible segments may reconnect. The resulting configuration has zero net winding and can be deformed into an unwound string. The board traces precisely this opposite-orientation reconnection.

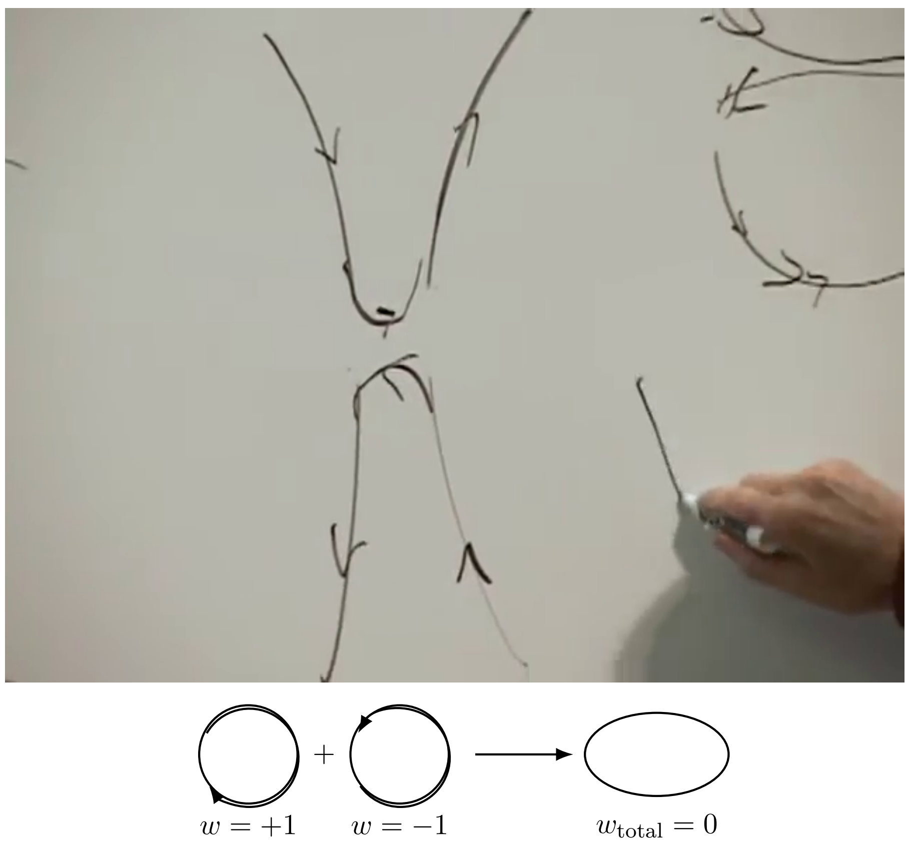

Figure 1.2: Opposite windings reconnect to a zero-net-winding configuration. The lower diagram is a clean reconstruction of the board move, not a second dynamical process.

Lecture frame: 00:13:20.

Now repeat the experiment with two strings having the same orientation. They can join, but they cannot unwind. The result carries twice the winding of either incoming string. The number of connected strings has changed; the total winding has not.

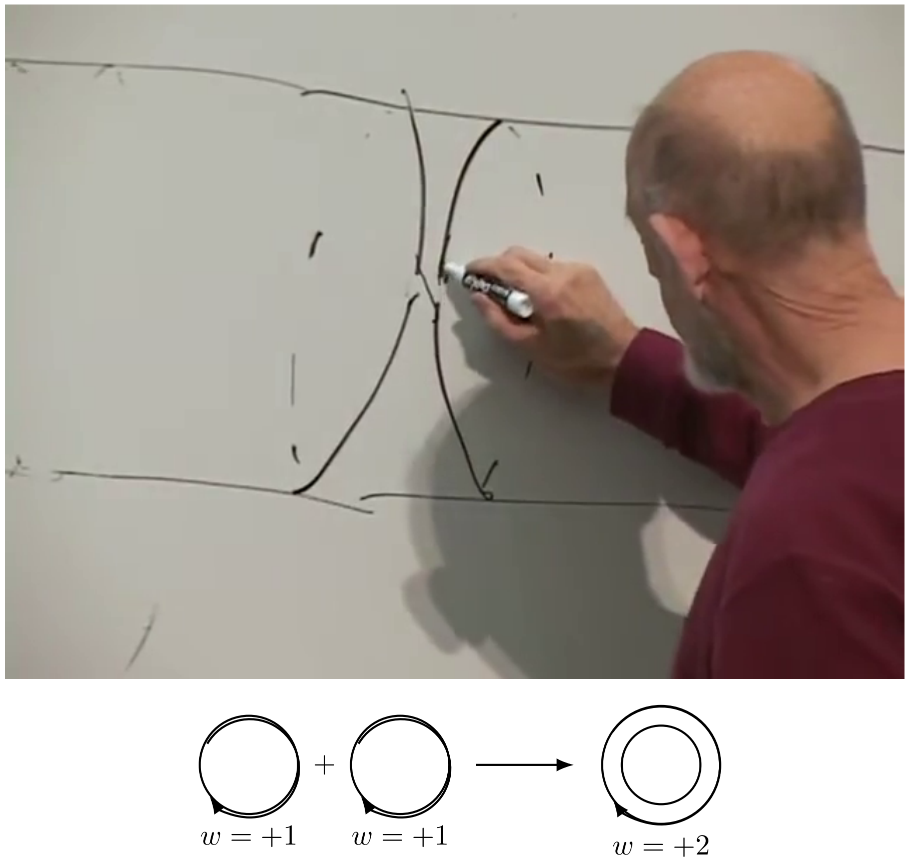

Figure 1.3: Same-sign windings can reconnect into a doubly wound string. Connectivity changes while total winding remains two.

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This distinction survives every allowed process. A total winding of one hundred might be carried by one string wound one hundred times, two strings wound fifty times each, or one hundred singly wound strings. Strings can split and join among these descriptions, but the signed sum remains one hundred. We denote it by $w \in \mathbb{Z}$.

1.3 Momentum Modes and Winding Modes

Conservation by itself is not yet duality. We next compare energies. A state moving around the compact circle has quantized momentum

$$p_Y = \frac{n}{R}, \qquad n \in \mathbb{Z}. \tag{1.2}$$

For a state whose other mass contributions can be neglected, the compact motion contributes $E_{\mathrm{mom}} = |n|/R$. Positive and negative n denote opposite directions of motion, while the energy is positive.

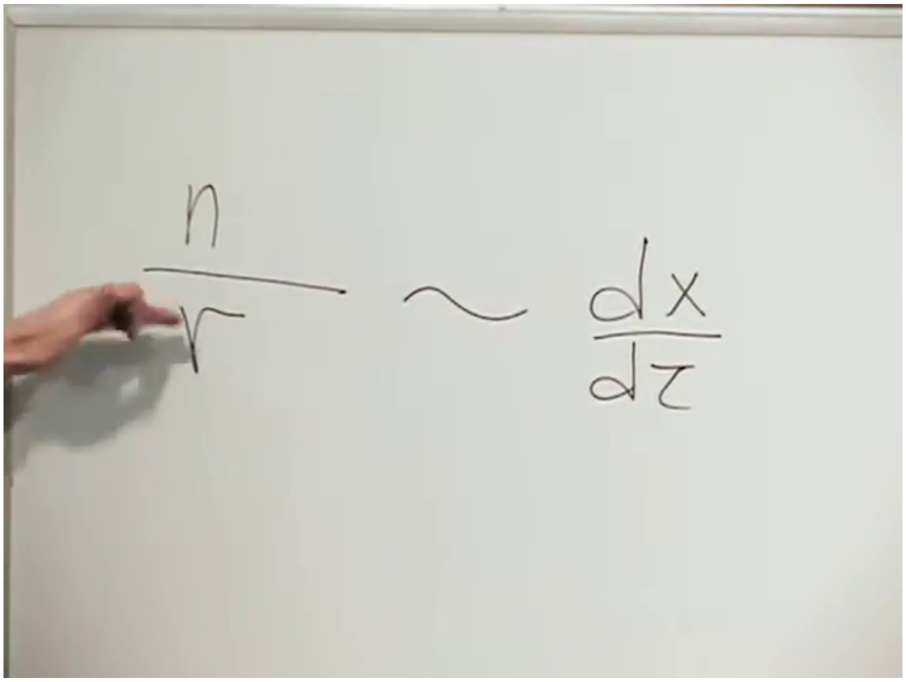


Figure 1.4: The compact momentum quantum number on the board. The lecture writes the momentum schematically as an integral of the compact velocity.

Lecture frame: 00:21:30.

A string wound $|w|$ times has length $2\pi|w|R$. With fundamental string tension

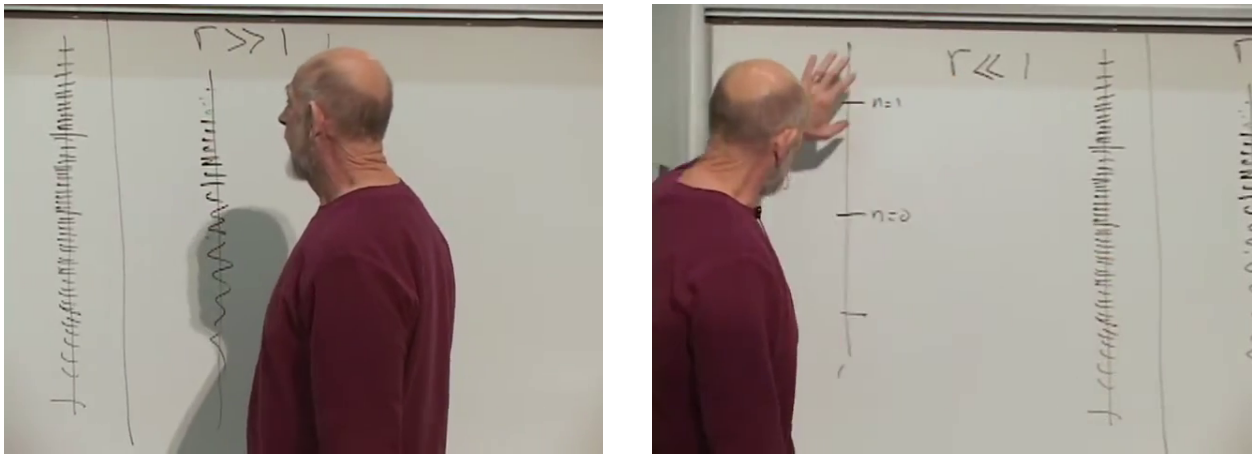
$$T_{\mathrm{F}} = \frac{1}{2\pi\alpha'}, \tag{1.3}$$

its winding energy is

$$E_{\mathrm{wind}} = T_{\mathrm{F}}(2\pi|w|R) = \frac{|w|R}{\alpha'}. \tag{1.4}$$

In the lecture units $\alpha' = 1$, this is simply $E_{\mathrm{wind}} = |w|R$.

The two towers behave oppositely as the radius changes. For $R \gg \sqrt{\alpha'}$, momentum levels are closely spaced and winding levels are widely spaced. For $R \ll \sqrt{\alpha'}$, momentum levels are widely spaced and winding levels are closely spaced.



Large compact radius. *Small compact radius.*

Figure 1.5: The two spectral comparisons on the board. Sparse and dense towers exchange roles as the radius is inverted.

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The clean schematic is

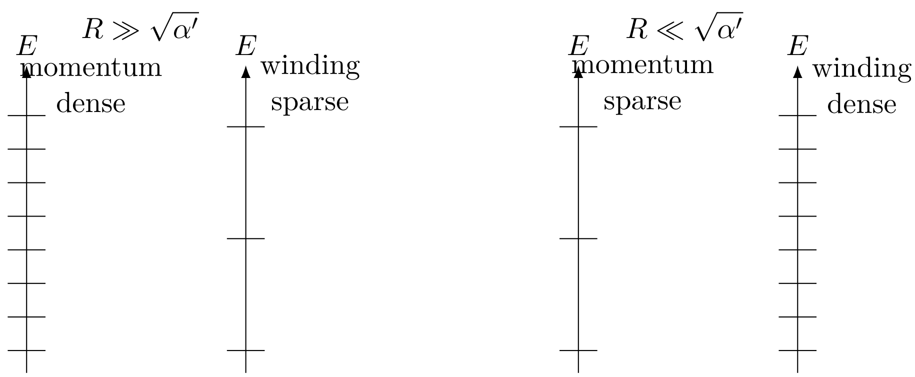


Figure 1.6: Reconstruction of the level-spacing argument. The complete string spectrum is unchanged when momentum and winding are exchanged together with $R \leftrightarrow \alpha'/R$.

Suppose an observer is handed one of these spectra without being told how it was made. Is the closely spaced tower momentum on a large circle, or winding on a small circle? The spectrum alone cannot say. String theory makes the stronger statement that scattering amplitudes cannot say either. The two descriptions are exactly equivalent under

$$n \longleftrightarrow w, \qquad R \longleftrightarrow \widetilde{R} = \frac{\alpha'}{R}. \tag{1.5}$$

The fixed point is

$$R = \sqrt{\alpha'}, \tag{1.6}$$

or $R = 1$ in the lecture units. At that radius additional states can become massless and the gauge symmetry is enhanced.

Question from the audience. Is winding only a mathematical convenience for rewriting the spectrum? [a]

Response. No. Winding labels real sectors of the closed-string Hilbert space, and those sectors take part in interactions. The word "duality" means that the momentum description and winding description are different coordinates on the same physics, not that one sector is an optional mnemonic for the other.

[a]Lecture timestamp: 00:27:33.

Question from the audience. Is there a radius that is mapped to itself? [a]

Response. Yes. The self-dual value is $R = \sqrt{\alpha'}$, which is $R = 1$ in the units used on the board. The two towers meet there, and the spectrum has enhanced symmetry.

[a]Lecture timestamp: 00:27:58.

1.4 The Worldsheet Form of the Duality

The spectral argument is suggestive, but the worldsheet makes the exchange more precise. In conformal gauge, the momentum conjugate to Y is proportional to $\partial_\tau Y$. Thus the total compact momentum has the form

$$P_Y = \frac{1}{2\pi\alpha'} \int_0^{2\pi} d\sigma\, \partial_\tau Y. \tag{1.7}$$

The winding number is

$$w = \frac{1}{2\pi R} \int_0^{2\pi} d\sigma\, \partial_\sigma Y = \frac{Y(2\pi) - Y(0)}{2\pi R}. \tag{1.8}$$

Question from the audience. Why does the integral of a derivative not vanish around the closed string? [a]

Response. Because Y is a periodic target-space coordinate, not a single-valued real function on the σ-circle. The worldsheet point $\sigma = 2\pi$ is the same as $\sigma = 0$, but a lift of the embedding to the covering line changes by $2\pi w R$. The integral of $\partial_\sigma Y$ measures that shift.

[a]Lecture timestamp: 00:32:33.

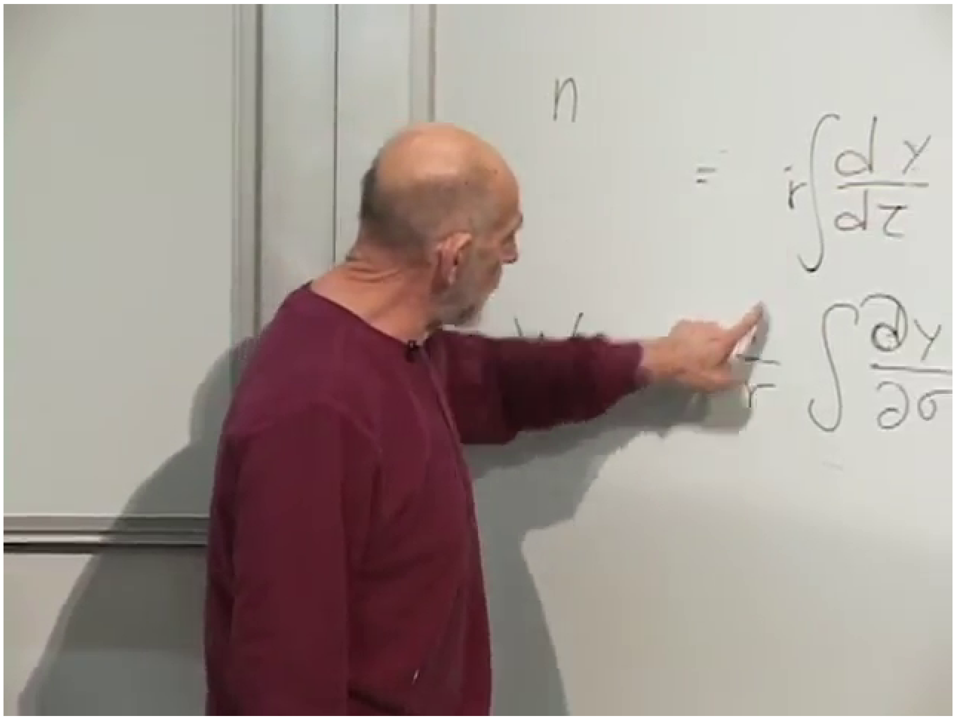


Figure 1.7: The momentum and winding integrals after both lines have been completed on the board. Common 2π and α' factors are suppressed in the lecture normalization.

Lecture frame: 00:33:25.

For a once-wound string,

$$Y(\tau, \sigma + 2\pi) = Y(\tau, \sigma) + 2\pi R. \tag{1.9}$$

For winding w, the right-hand shift is $2\pi w R$. Equation (1.8) then counts the winding exactly. This is the precise version of the board shorthand $w \sim R^{-1} \int d\sigma\, \partial_\sigma Y$.

The worldsheet statement of T-duality is therefore

$$\partial_\tau Y \longleftrightarrow \partial_\sigma \widetilde{Y}, \qquad \partial_\sigma Y \longleftrightarrow \partial_\tau \widetilde{Y}, \tag{1.10}$$

with momentum in Y becoming winding in the dual coordinate $\widetilde{Y}$, and conversely.

Question from the audience. Does this mean that the worldsheet coordinates τ and σ themselves are related or interchanged? [a]

Response. No. T-duality maps solutions and exchanges particular derivatives of the compact target coordinate. It does not identify the worldsheet time coordinate with the worldsheet spatial coordinate.

[a] Lecture timestamp: 00:34:39.

1.5 Why Winding States Cannot Be Avoided

One might ask why a string would ever be found in a wound state when an unwound state is available. A string does not choose its state by preference. Energy supplied in a collision stretches it.

A sufficiently energetic, highly distorted string can reconnect into two strings with opposite winding. Their total winding remains zero, but after the split the two strings may separate. One member of the pair can be sent far away, leaving an observer with an apparently isolated wound string.

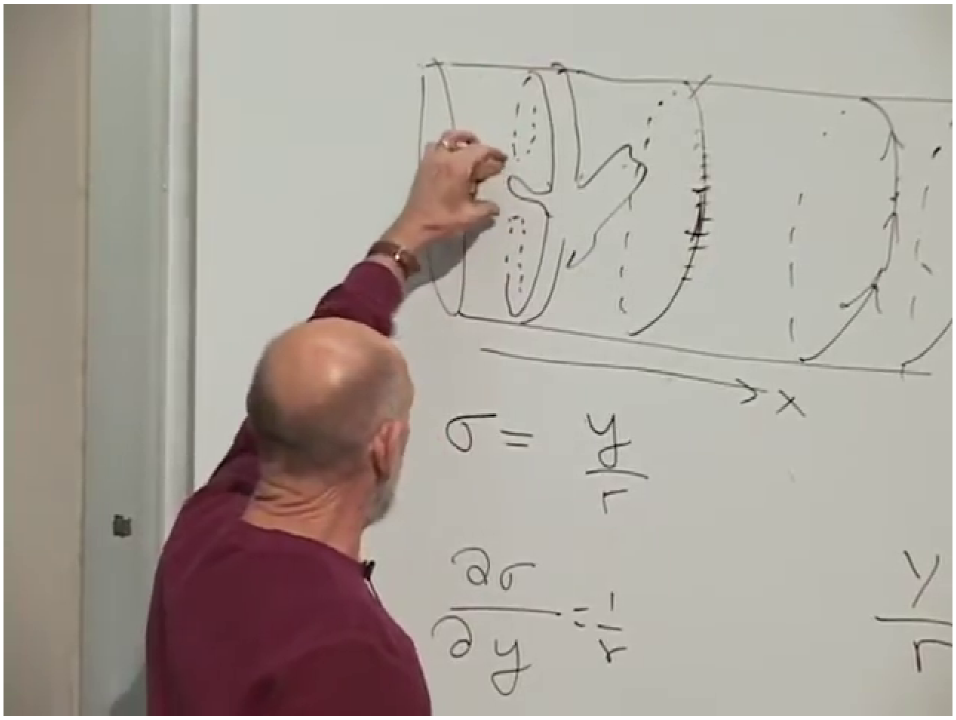


Figure 1.8: A highly stretched zero-net-winding configuration from which oppositely wound strings can separate after reconnection.

Lecture frame: 00:36:15.

The energy threshold is set by the winding energy. On a large circle it takes a large collision energy to stretch enough string around the compact direction. Above threshold, however, winding–antiwinding pair production cannot be forbidden. The analogy is photon collisions producing an electron–positron pair. The total electric charge is still zero, yet the charged particles are present and can be separated. Winding conservation works in the same way.

This analogy is more than verbal. Compact momentum and winding behave as two distinct charge quantum numbers. Like signs repel and opposite signs attract in the corresponding reduced description. We should therefore ask for the two vector fields that mediate those forces.

1.6 Momentum Charge from Five-Dimensional Gravity

Begin with gravity in five dimensions. Let capital indices M, N run over the four ordinary spacetime directions and the compact fifth direction. The metric decomposes as

$$G_{MN} quad \longrightarrow \quad \{G_{\mu\nu},, G_{\mu 5},, G_{55}\}. \tag{1.11}$$

The symmetric component $G_{5\mu}$ contains no new information beyond $G_{\mu 5}$.

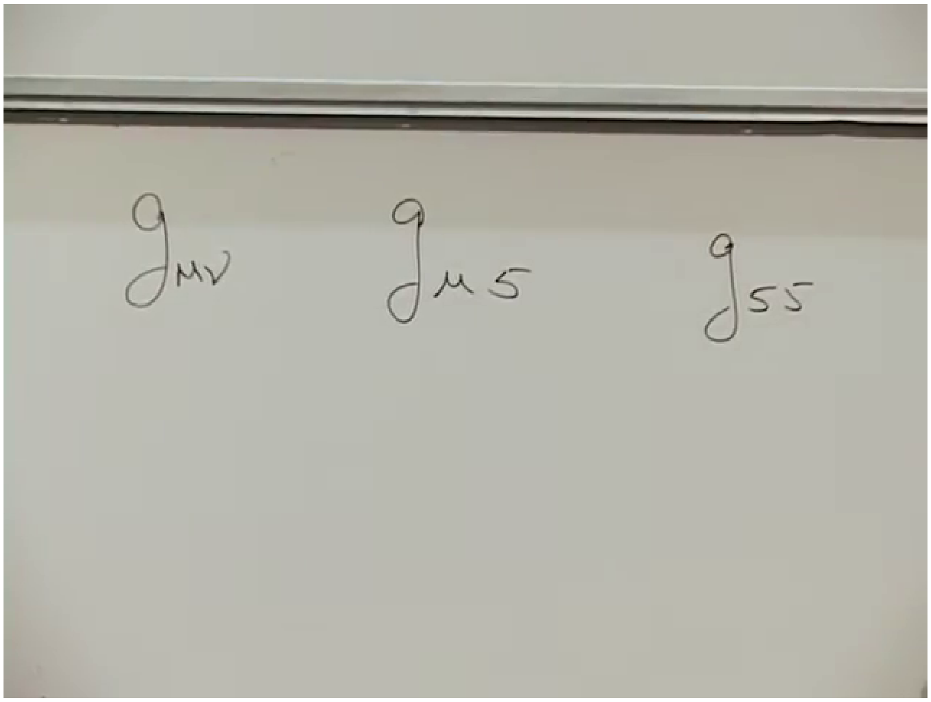


Figure 1.9: The independent groups of five-dimensional metric components: $G_{\mu\nu}$, $G_{\mu 5}$, and G_{55}.
Lecture frame: 00:44:05.

From the four-dimensional point of view,

$$G_{\mu\nu} \quad \text{is a spacetime metric,} \;\; G_{\mu 5} \quad \equiv A_\mu \quad \text{is a vector field,} \;\; G_{55} \equiv e^{2\varphi} \quad \text{is a scalar modulus.} \tag{1.12}$$

The scalar φ measures the size of the compact direction. It can vary smoothly over ordinary spacetime, so waves of φ are waves in the radius of the fifth dimension.

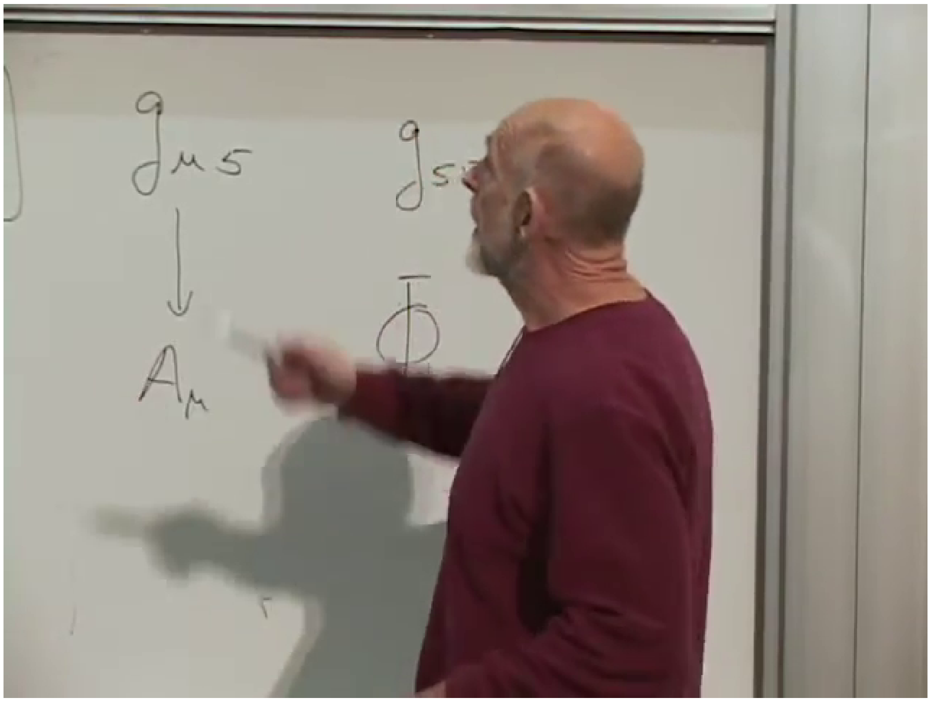


Figure 1.10: The board identifies $G_{\mu 5}$ with a vector A_μ and G_{55} with a scalar Φ.
Lecture frame: 00:47:40.

The source of a mixed gravitational component is mixed momentum flow. In particular, momentum around the fifth direction sources $G_{\mu 5}$. A four-dimensional observer therefore interprets the integer n as an electric charge coupled to A_μ. This is the Kaluza–Klein mechanism: what is momentum in the higher-dimensional geometry becomes charge in the lower-dimensional field theory.

[1]

Question from the audience. Is the scalar φ proportional to the momentum integer n? [a]

Response. No. The integer n labels a quantized state and acts as charge. The scalar $\varphi(x)$ is a continuous field that determines the local compactification size. It may vary from place to place; it is not itself an integer.

[a] Lecture timestamp: 00:48:43.

1.7 The Second Vector Field and Winding Charge

Winding is also charge-like, but it cannot couple to the same vector in the same way. To see the second vector, return to the first closed-string excitations. A closed string has left-moving and right-moving oscillator modes. Physical states obey level matching: the total left-moving and right-moving excitation levels must agree.

In the bosonic theory the unexcited state is the tachyon. A single left-moving or right-moving oscillator would violate level matching. The first allowed massless state therefore contains one

[1] Editorial clarification: The lecture loosely calls the scalar from G_{55} a "dilaton." In this reduction it is specifically the radius modulus. The ten-dimensional string dilaton, which controls the string coupling, is a separate field, although lower-dimensional scalar combinations can mix under changes of frame and duality.

oscillator of each kind,

$$\alpha^I_{-1}\tilde{\alpha}^J_{-1}\,|0;k\rangle\,. \tag{1.13}$$

The indices I, J run over transverse spatial directions, including the compact direction. If one index is ordinary and the other is 5, there are two vector-like oscillator states,

$$\alpha^\mu_{-1}\tilde{\alpha}^5_{-1}\,|0;k\rangle\,, \qquad \alpha^5_{-1}\tilde{\alpha}^\mu_{-1}\,|0;k\rangle\,. \tag{1.14}$$

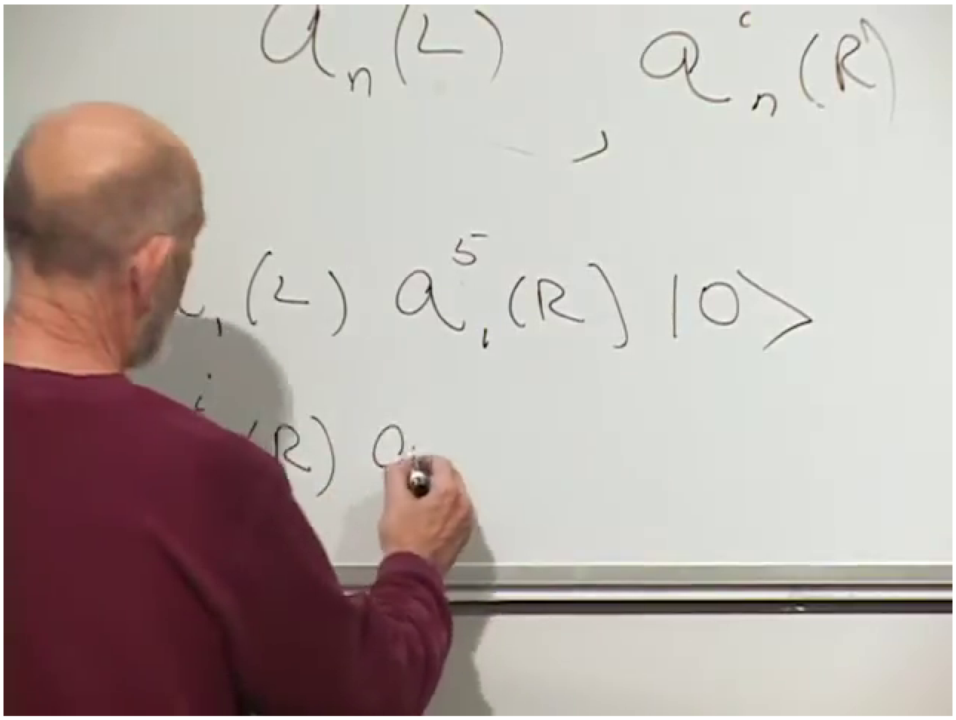


Figure 1.11: The two left–right oscillator orderings that become four-dimensional vector states when one polarization points along the compact direction.

Lecture frame: 00:58:20.

Their symmetric and antisymmetric combinations belong to different spacetime fields:

$$\alpha^{(\mu}_{-1}\tilde{\alpha}^{5)}_{-1}\,|0;k\rangle \longleftrightarrow G_{\mu 5},\ \ \alpha^{[\mu}_{-1}\tilde{\alpha}^{5]}_{-1}\,|0;k\rangle \qquad\qquad \longleftrightarrow B_{\mu 5}. \tag{1.15}$$

The first is the Kaluza–Klein vector sourced by compact momentum. The second descends from the antisymmetric Kalb–Ramond field and is sourced by string winding. Thus the reduced theory contains two photon-like fields, one for each charge sector.

If both oscillator indices point along the compact direction, the state is a scalar associated with the compactification modulus. If both are in ordinary transverse directions, the symmetric traceless part is the graviton, the antisymmetric part is $B_{\mu\nu}$, and the trace contributes to a scalar sector.

Question from the audience. Which of these states is the singlet and which is the triplet? [a]

Response. That classification is not the useful one here. Both mixed states transform as four-dimensional vectors. Their sum and difference separate the symmetric metric polarization from the antisymmetric B-field polarization. The state with two compact indices is instead a four-dimensional scalar.

[a] Lecture timestamp: 01:00:21.

The complete dictionary reached before the break is

$$\boxed{n \leftrightarrow w, qquadR \leftrightarrow \frac{\alpha'}{R}, qquad\partial_\tau Y \leftrightarrow \partial_\sigma \widetilde{Y}, qquadG_{\mu 5} \leftrightarrow B_{\mu 5}.} \tag{1.16}$$

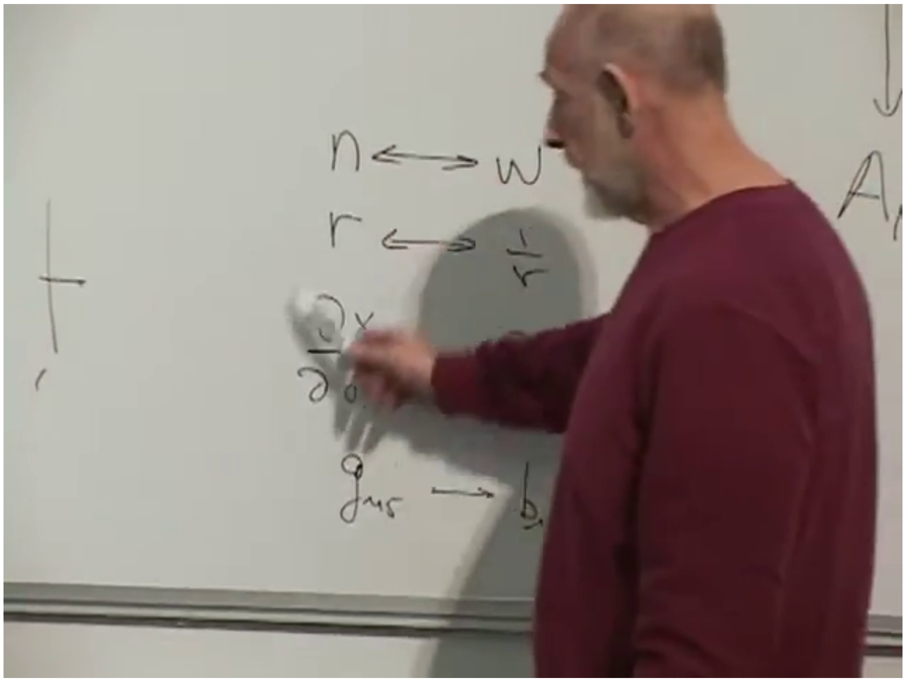


Figure 1.12: The board summary of the T-duality map: momentum and winding, radius and inverse radius, worldsheet derivatives, and the two vector fields are exchanged. The lecturer partly obscures the final line, which is therefore supplied explicitly in Equation (1.16).

Lecture frame: 01:02:20.

This construction should not be confused with a claim that every field just listed has already been identified in the observed particle spectrum. At this point it is a mathematical consequence of the compactified string theory. There are many proposals for connecting such structures to phenomenology, but that is not the problem being solved in this lecture.

1.8 Open Strings Change the Question

After the break, we begin again with open strings. A closed string can have stable winding because it has no endpoint through which the winding can slip away. An open string has endpoints and therefore no analogous topologically protected winding number. Yet T-duality is an exact symmetry of the theory. What does the symmetry do to an open string?

For a freely moving endpoint, the variation of the string action gives a Neumann boundary condition. In the simple (X, Y) geometry,

$$\partial_\sigma X|_{\partial\Sigma} = 0, \qquad \partial_\sigma Y|_{\partial\Sigma} = 0. \tag{1.17}$$

The mechanical picture is straightforward. Regard the string as a chain of ever smaller mass elements. A nonzero tangent force on the last element would give it an acceleration that diverges as its mass tends to zero. A free endpoint therefore adjusts until the net endpoint tension vanishes. Someone could instead hold the endpoint fixed and supply the necessary force, but then the endpoint would no longer be free.

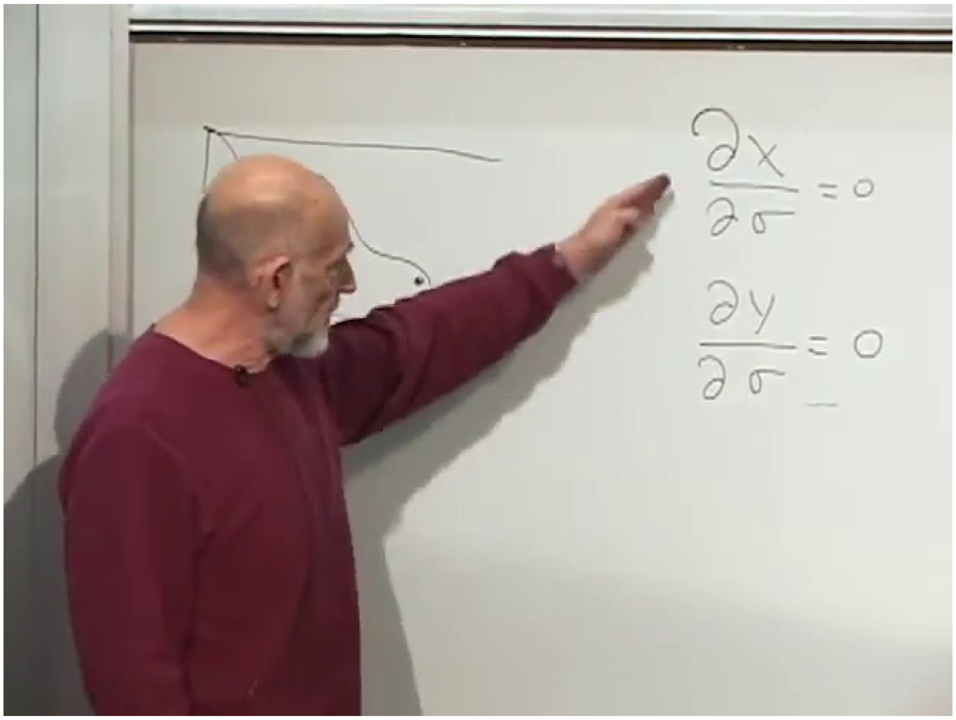


Figure 1.13: The open string and its completed Neumann endpoint conditions, $\partial_\sigma X = \partial_\sigma Y = 0$.
Lecture frame: 01:09:20.

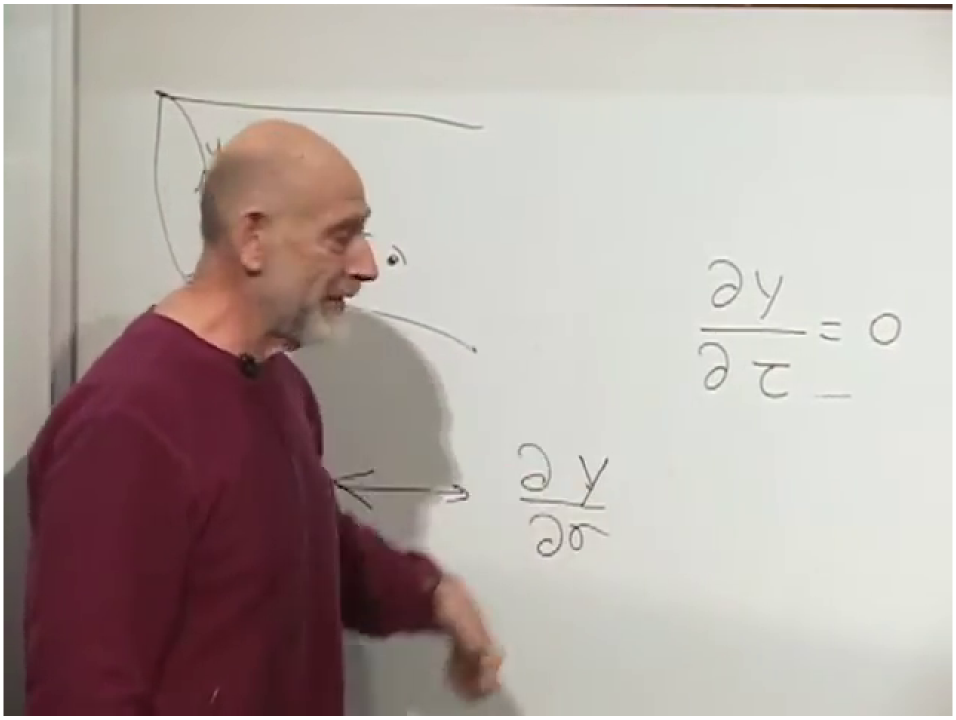
Now dualize only the compact coordinate Y. The noncompact coordinate X is unchanged. Equation (1.10) turns the second Neumann condition into

$$\partial_\tau \widetilde{Y}\Big|_{\partial\Sigma} = 0. \tag{1.18}$$

Therefore the endpoint has a time-independent position in the dual compact direction,

$$\widetilde{Y}\Big|_{\partial\Sigma} = Y_0. \tag{1.19}$$

This is a Dirichlet boundary condition.



$$\begin{array}{ccc} \text{Neumann} & & \text{Dirichlet} \\ \partial_\sigma Y|_{\partial\Sigma} = 0 & \xrightarrow{T_Y} & \partial_\tau \widetilde{Y}\Big|_{\partial\Sigma} = 0 \end{array}$$

$$\widetilde{Y}(\tau) = Y_0 \text{ at each endpoint}$$

Figure 1.14: The completed dual endpoint condition on the board and its clean logical reconstruction. T-duality turns freedom in the compact direction into a fixed endpoint position in the dual direction.

Lecture frame: 01:12:35.

Question from the audience. Why does the dual condition mean that the endpoint is fixed? [a]

Response. Because $\partial_\tau \widetilde{Y} = 0$ is zero velocity in the dualized direction. Wherever the endpoint begins in $\widetilde{Y}$, it remains there. Its coordinates parallel to the un-dualized directions are still free.

[a] Lecture timestamp: 01:12:39.

1.9 The Object Holding the Endpoint Is a D-Brane

If an endpoint is fixed, something must be able to hold it there. Exact T-duality therefore requires a new object on which open strings can end. That object is a D-brane. The letter *D* stands for Dirichlet, not for the dimension of the object. In the historical aside of the lecture, the objects might have been called after Joseph Polchinski, whose analysis made their dynamical role unmistakable, but the already established term "p-brane" left Dirichlet as the natural name.

A p-brane has p spatial dimensions. A string is a one-brane, a membrane is a two-brane, and an extended three-dimensional object is a three-brane. A Dp-brane is a p-brane on which fundamental-string endpoints obey Dirichlet conditions in the transverse directions and Neumann conditions in the directions tangent to the brane.

Nothing in the duality argument singles out one value of Y_0, so the D-brane cannot be a rigid fixture nailed to a preferred point in the universe. It must be movable. Relativity also forbids an absolutely rigid extended body, so a D-brane can fluctuate and bend. The simple flat brane drawn on the board is its equilibrium configuration, not a restriction on its dynamics.

Question from the audience. Must the two endpoints of an open string have the same fixed value of the dual coordinate? [a]

Response. Not in general. Both endpoints may lie on the same D-brane, in which case they share that brane's transverse location. They may also lie on two different D-branes at different transverse positions, and the string then stretches between them. A brane itself may fluctuate and bend, so the endpoint locus need not remain a perfectly flat plane.

[a]Lecture timestamp: 01:14:55.

1.10 Counting the Dimensions of a D-Brane

Ten-dimensional superstring theory has nine spatial dimensions. Suppose T-duality imposes a Dirichlet condition in one spatial direction. The endpoint is still free in the other eight, so its allowed locus is an eight-dimensional D8-brane. Each additional independent Dirichlet condition removes one direction of endpoint motion. If k spatial coordinates are fixed, then

$$p = 9 - k. \tag{1.20}$$

The tabletop gives the three-dimensional version of the same count. Fix the vertical coordinate and a fingertip may move on a two-dimensional tabletop. Fix a second coordinate and it may move only along a line. Fix all three and it is confined to a point.

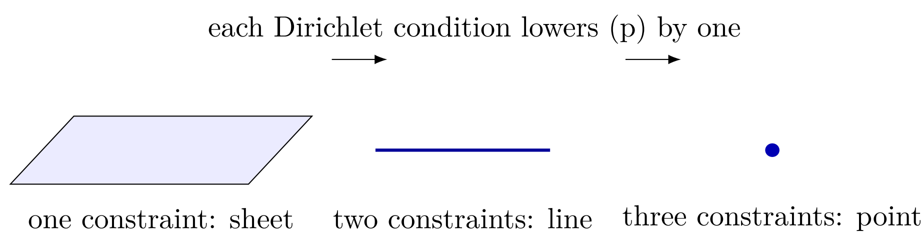


Figure 1.15: The dimensional-counting argument used at the tabletop. A Dp-brane is the locus in which endpoints remain free to move.

Repeated dualities generate D8-, D7-, D6-branes and so on down to a D0-brane, which is pointlike in space. If no spatial direction is fixed, a D9-brane fills all nine spatial dimensions and ordinary open strings are free to move everywhere. The lecture briefly mentions D(-1)-branes; these are localized in Euclidean spacetime and behave as instantons rather than as ordinary particles propagating through time.
[2]

[2]Editorial clarification: The dimensional ladder is a statement about the type-II theories taken together. Stable

Question from the audience. If only six dimensions are compact, should the counting start by subtracting from six rather than nine? [a]

Response. The elementary derivation performs T-duality on compact directions, but a dual circle becomes arbitrarily large as the original circle becomes arbitrarily small. In that limit it is effectively noncompact. The resulting branes can therefore extend along or be transverse to directions that appear noncompact at accessible scales. The dimension p counts all spatial directions tangent to the brane, not only the compact ones used in one derivation.

[a]Lecture timestamp: 01:20:49.

A D0-brane is a new kind of particle. An open string may run from one D0-brane to another, return to the same D0-brane, or stretch toward a very distant brane. A string taken literally to spatial infinity has infinite length and hence infinite energy, so the physically useful version places its other endpoint on another object.

The next case, a D1-brane, is itself string-shaped, but it is not the fundamental string with which we began. It can bend and close on itself, and fundamental strings may end on it. Its tension scales differently with the string coupling:

$$T_{\mathrm{D1}} = \frac{1}{2\pi\alpha' g_s}, \qquad T_{\mathrm{F1}} = \frac{1}{2\pi\alpha'}. \tag{1.21}$$

At weak coupling $g_s \ll 1$, the D-string is much heavier than the fundamental string, which explains why it first appears as a massive anchor.

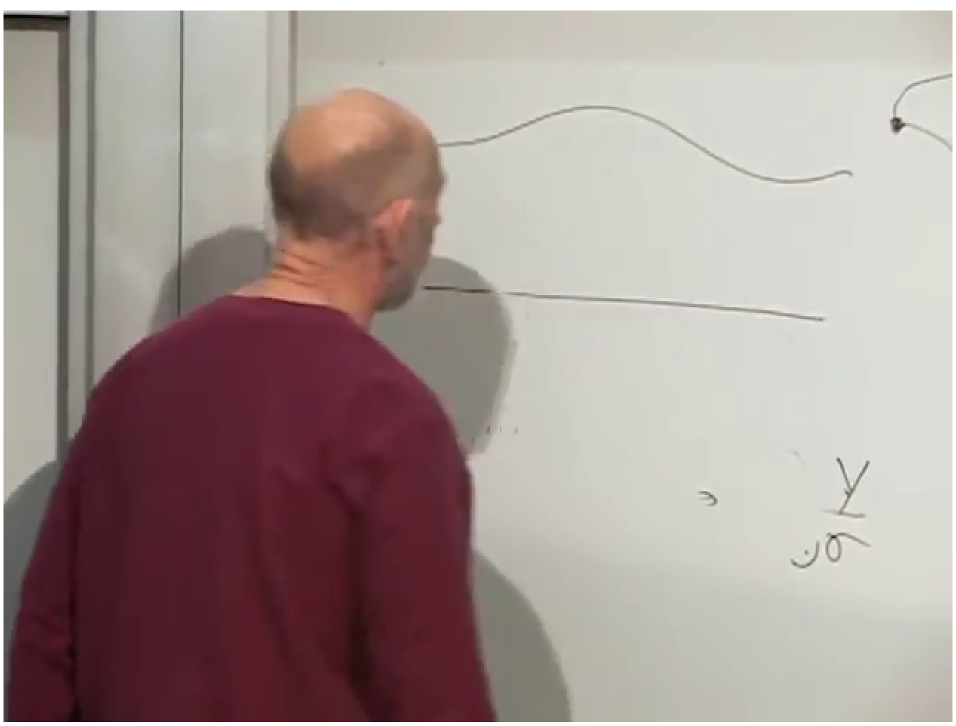


Figure 1.16: The board's D1-brane: a bendable one-dimensional object, distinct from the fundamental strings that may end on it.

Lecture frame: 01:24:20.

BPS Dp-branes have even p in type IIA and odd p in type IIB. T-duality along one circle changes p by one and simultaneously exchanges type IIA with type IIB.

Question from the audience. Does T-duality act on the D-branes as well as on the strings? [a]

Response. Yes. A T-duality parallel to a Dp-brane lowers its dimension to D$(p-1)$, while a T-duality transverse to it raises the dimension to D$(p+1)$. Branes are part of the duality dictionary, not external supports left unchanged by it.

[a] Lecture timestamp: 01:25:35.

Question from the audience. May directions tangent to a brane still be compact? [a]

Response. Yes. "Tangent" and "compact" answer different questions. A direction may lie along the brane and still close into a circle; it may also be noncompact. What matters for the endpoint is whether the boundary condition in that direction is Neumann or Dirichlet.

[a] Lecture timestamp: 01:27:25.

A D2-brane is a membrane-like sheet. A D3-brane has three spatial dimensions. In a genuinely higher-dimensional spacetime a D3-brane is a three-dimensional subspace rather than all of space, even though observers confined to it may experience it as their entire spatial world.

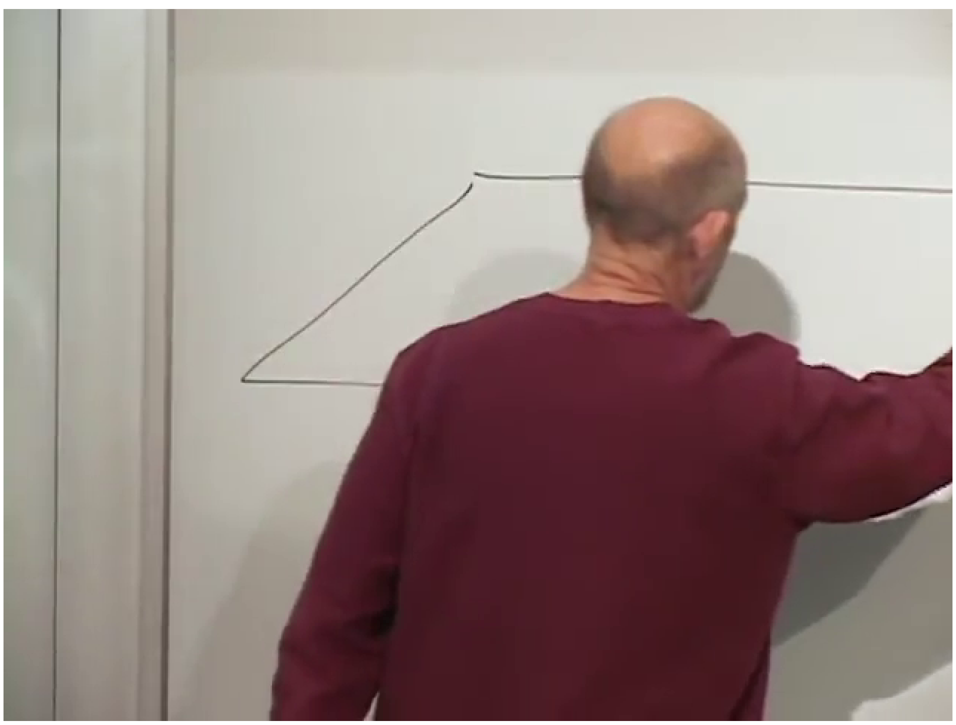


Figure 1.17: A D2-brane drawn as a sheet on which string endpoints can move.

Lecture frame: 01:27:55.

The derivation given here is intentionally a sketch. It is not by itself a proof that string theory describes nature. Within string theory, however, the consistency checks are numerous: dualities, quantum calculations, and mathematical results all point to D-branes as genuine dynamical objects.

1.11 Open Strings Become Fields on a Brane

Take an otherwise empty D2-brane and place short open strings on it. Each endpoint is confined to the sheet, so the whole low-energy excitation moves along the sheet. Two oriented strings whose adjacent endpoints carry compatible arrows can join. The joined string may later split again. Read as particle histories, these are the vertices of Feynman diagrams: two particles enter, an intermediate excitation propagates, and two particles emerge.

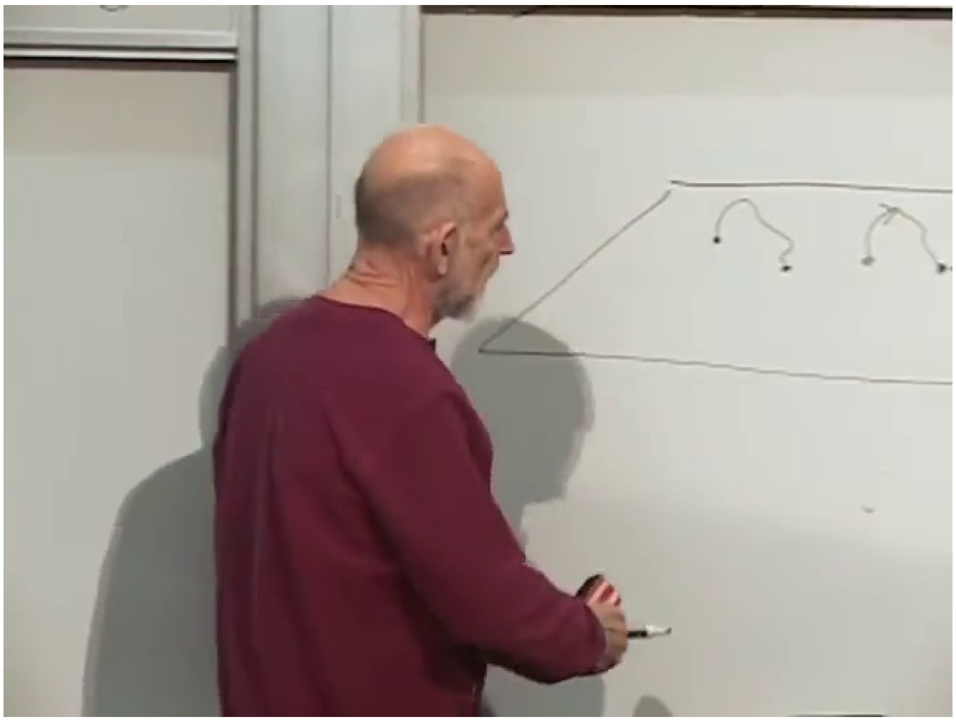


Figure 1.18: Oriented open strings on a D2-brane joining through compatible endpoints. At low energy the process is read as a field-theory interaction on the brane.

Lecture frame: 01:29:55.

At energies well below the string scale, the internal vibrational tower is not excited. The remaining massless open-string state is a vector field on the brane worldvolume. For one brane this is an abelian $U(1)$ gauge field, the brane analogue of electromagnetism. The field does not propagate through the full ten-dimensional bulk; its quanta are open strings whose endpoints keep them on the brane.

1.12 A Stack of Branes Produces Nonabelian Color

Now place several parallel branes near one another. For three branes, label them Red, Green, and Blue. An oriented string has an ordered pair of endpoint labels: it may run from Red to Green, Green to Blue, Red to Red, and so on. With N branes the massless open-string states can be arranged as an $N \times N$ matrix

$$A_\mu{}^i{}_j, \qquad i, j = 1, \ldots, N. \tag{1.22}$$

These are the Chan–Paton labels of a $U(N)$ gauge theory.

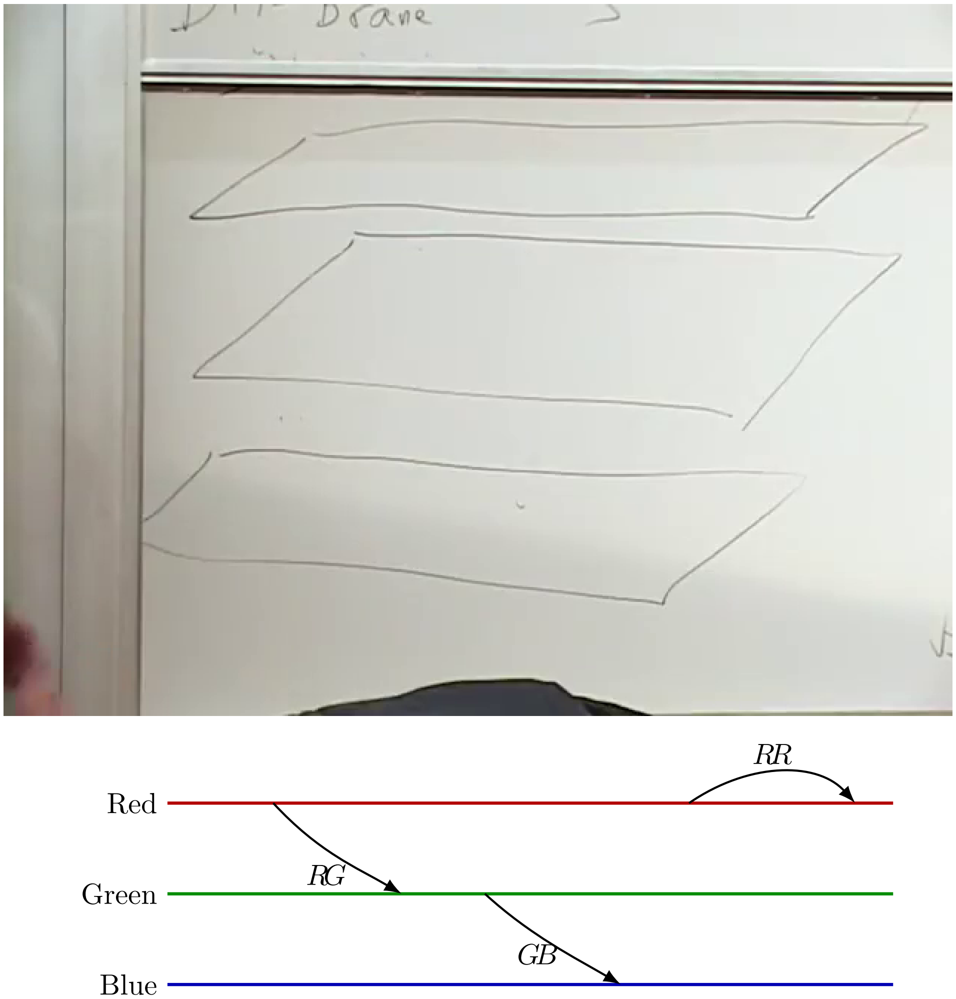


Figure 1.19: Three parallel branes on the board and a clean reconstruction of the ordered endpoint labels carried by strings between them.

Lecture frame: 01:33:15.

Question from the audience. Can a string really run from the Red brane to the Green brane without being blocked by the other drawings? [a]

Response. Yes. The branes are embedded in a higher-dimensional space. The two-dimensional board picture is only a projection, so a string may pass around another projected line in a transverse direction.

[a]Lecture timestamp: 01:35:07.

Question from the audience. Must a Red–Green string combine with a Green–Blue string in order to become Red–Blue? [a]

Response. That is the basic local joining rule. Compatible endpoints meet, and the two segments become one string whose unjoined endpoints retain their Red and Blue labels. More elaborate processes are built by repeating the same operation.

[a]Lecture timestamp: 01:35:39.

The joining rule is matrix multiplication made geometric. A Red–Green string can join a Green–Blue string because the Green endpoint of one is compatible with the Green endpoint of the other. The result is a Red–Blue string:

$$(R, G) \circ (G, B) \longrightarrow (R, B). \tag{1.23}$$

The repeated index is joined and disappears internally. This is the same index structure that appears at a Yang–Mills interaction vertex.

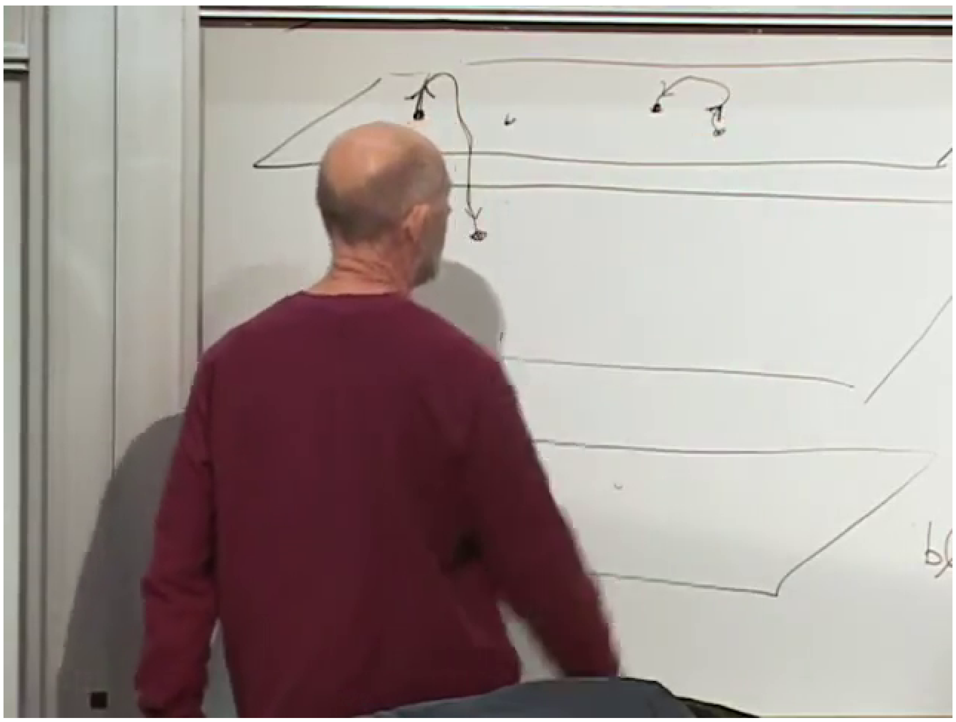


Figure 1.20: Joining open strings on a stack. Endpoint orientation enforces the same ordered-index composition used by nonabelian gauge fields.

Lecture frame: 01:36:30.

For three coincident branes the complete open-string gauge group is $U(3)$, with nine vector states. Its algebra decomposes as

$$\mathfrak{u}(3) = \mathfrak{su}(3) \oplus \mathfrak{u}(1). \tag{1.24}$$

The eight traceless combinations have the color structure of the QCD gluons, while the diagonal center-of-mass $U(1)$ is a separate abelian field.

[3]

Quark-like matter carries only one color index. In the brane picture it is represented by a string with one endpoint on the color stack and the other on a different, distant brane. Relative to the color stack, only one end is visible as a color label. Reversing the orientation gives the antiparticle.

[3]Editorial clarification: The lecture describes one of the nine color-pair combinations as able to detach and disappear, thereby leaving eight gluons. The precise D-brane statement is that a stack of three coincident branes gives $U(3)$, not only $SU(3)$. Obtaining the QCD octet requires separating, projecting out, or otherwise decoupling the diagonal $U(1)$. The lecture's picture correctly motivates the ordered color indices, but it is not a complete derivation of QCD.

A quark-like and antiquark-like string with compatible endpoints can join and leave the color stack; two identically oriented quark-like strings cannot annihilate in that way.

The resulting low-energy theories are generally supersymmetric Yang–Mills theories, with additional matter determined by the branes and their intersections. They are close enough to QCD to illuminate nonabelian dynamics, but they are not automatically the Standard Model. With only one brane, the same construction reduces to a $U(1)$ theory: the short brane–brane strings are photon-like, while oriented strings reaching a second sector play the roles of positive and negative electric charges.

1.13 D-Strings End as Magnetic Monopoles

There are now two kinds of string-like object in the story. The original fundamental string, or F-string, carries electric flux on a D3-brane. The D1-brane, or D-string, is much heavier at weak coupling. A D-string can also end on a D3-brane. Its endpoint is not electrically charged under the worldvolume gauge field; it is magnetically charged. From the perspective of observers on the D3-brane, the endpoint is a magnetic monopole.

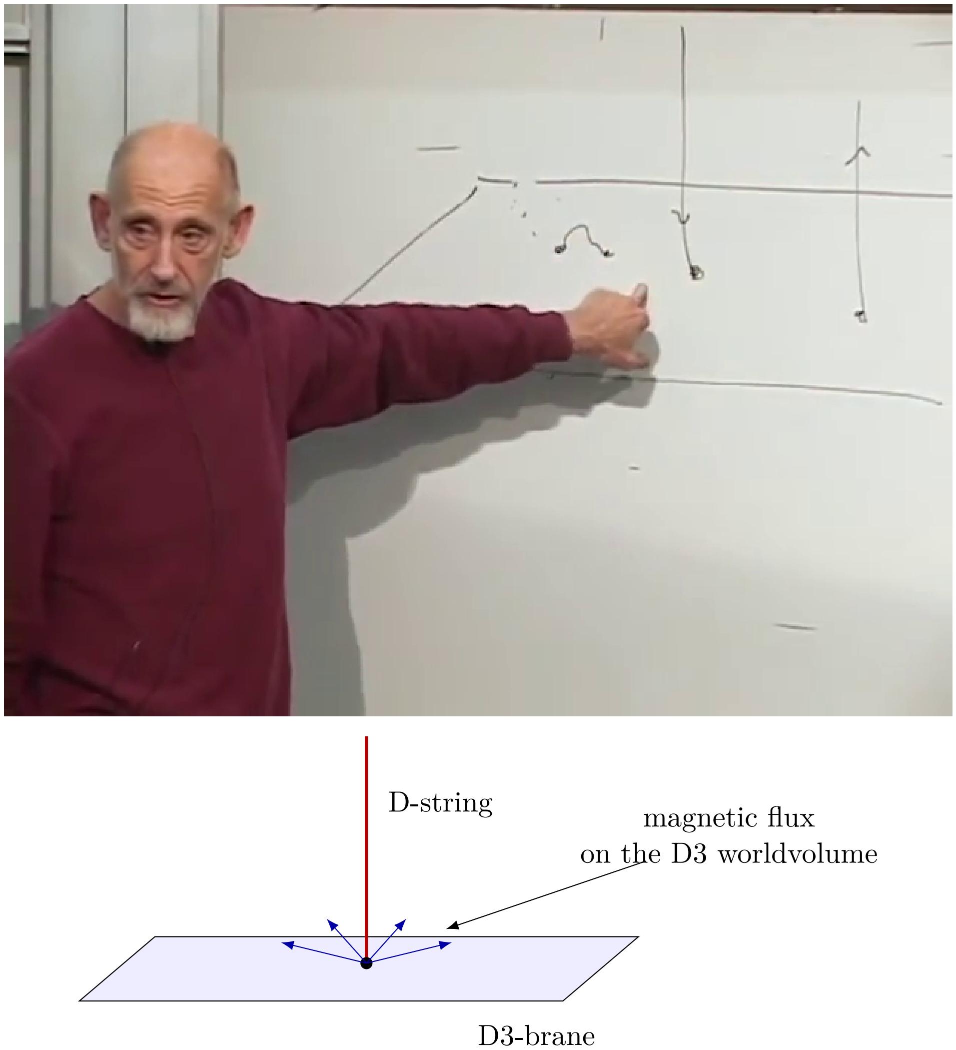


Figure 1.21: The board endpoint construction and its clean interpretation. A D-string ending on a D3-brane appears as a magnetic monopole to the worldvolume gauge theory.

Lecture frame: 01:41:30.

The mass hierarchy also has the right qualitative form. Since the D-string tension carries a factor $1/g_s$, its endpoint is much heavier than an ordinary electric excitation at weak coupling. The electric/magnetic relation between F-string and D-string endpoints is part of the wider duality structure of type IIB string theory.

Question from the audience. Is the endpoint of the D-string neutral? [a]

Response. It is neutral with respect to the electric charge carried by a fundamental-string endpoint, but it is not uncharged. It carries magnetic charge and is therefore a monopole in the D3-brane gauge theory.

[a] Lecture timestamp: 01:42:19.

1.14 What the Construction Does and Does Not Claim

The path from T-duality to branes was assembled over decades, not in a single calculation. A theory first proposed to describe hadrons became a candidate theory of quantum gravity. Its dualities then forced D-branes into the spectrum, and gauge theories reappeared as the low-energy dynamics of strings ending on those branes. In that sense the subject made a full circle.

D-brane and related dual constructions have become powerful tools for studying supersymmetric gauge theories, strongly coupled systems, and connections between gravity and field theory. They have also informed models of fluids and plasmas through gauge/gravity duality. This success is not the same as deriving the observed particle spectrum directly. The compactification geometry, brane configuration, supersymmetry breaking, and many other ingredients still have to be selected.

Question from the audience. What determines which compact dimension is dualized? [a]

Response. Mathematically, one may dualize any compact direction for which the required symmetry is present, choosing the description that is most convenient. Physically, the brane orientations and compactification data realized in a particular state are selected by the history and dynamics of that state.

[a] Lecture timestamp: 01:44:59.

Question from the audience. Which brane supports the monopole construction? [a]

Response. The world experienced by the gauge theory is a D3-brane. The D-string extends into a transverse direction and ends on that D3-brane; its endpoint is the monopole seen in the three spatial dimensions of the brane.

[a] Lecture timestamp: 01:45:21.

Question from the audience. Are D-strings oriented as well? [a]

Response. Yes. Reversing a D-string's orientation reverses the corresponding charge. The two endpoint orientations distinguish monopole from antimonopole just as orientation distinguishes electric charge for the fundamental-string endpoints.

[a] Lecture timestamp: 01:45:55.

Question from the audience. Can this simple toroidal picture account for the left–right asymmetry of chiral particle physics? [a]

Response. Not by itself. A flat torus is too symmetric. Chiral spectra require more structure, such as asymmetric brane intersections, orbifold projections, or compactification on suitable Calabi–Yau geometries. String theory has mechanisms for chirality, but they lie beyond this elementary T-duality construction.

[a] Lecture timestamp: 01:46:05.

We have followed one symmetry through four increasingly concrete forms: an exchange of spectra, an exchange of worldsheet derivatives, an exchange of spacetime fields, and finally an exchange of endpoint boundary conditions. The last form is the decisive one. It turns a mathematical equivalence into the discovery of D-branes, and it turns the dynamics of open strings into gauge theory on their worldvolumes. At that point the lecture has used up, as Susskind puts it, its time, energy, and momentum.